TEAM NAME: BandhuAi

PROBLEM STATEMENT: Privacy laws lock cancer data inside hospitals, forcing AI to train on small, inaccurate datasets

PROJECT NAME: MEDMASH
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PROBLEM STATEMENT

Collaborative cancer detection across hospitals without pooling or exposing sensitive patient health records

Institutional Data Silos

Hospitals maintain isolated clinical repositories, preventing oncology models from training on statistically adequate, multi-cohort populations.

Strict Regulatory Mandates

HIPAA, GDPR, and DISHA regulations strictly prohibit pooling or centralizing raw patient data across institutional boundaries.

Underperforming Local Models

Models trained exclusively on limited, localized cohorts fail to generalize and struggle to accurately classify rare malignancy variants.

Gradient Inversion Vulnerabilities

Exchanging model updates without calibrated Differential Privacy leaves distributed networks exposed to reverse-engineering and data-reconstruction attacks.

SOLUTION

Privacy-preserving collaborative oncology powered by Federated Learning and Differential Privacy

A

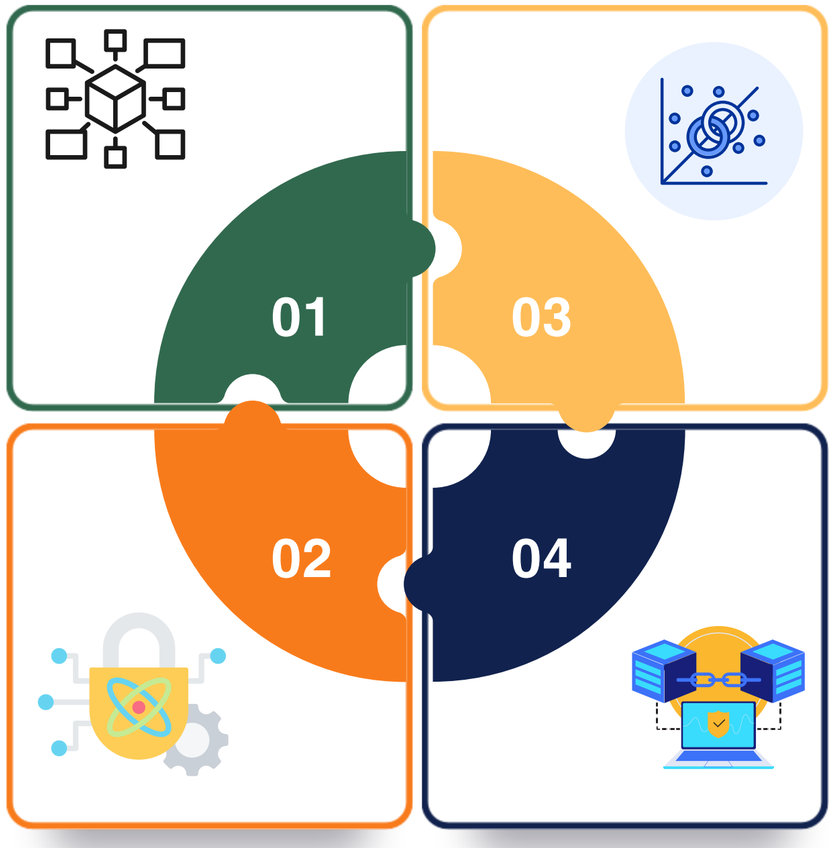
Federated Architecture

Diagnostic models train locally inside hospital firewalls without transmitting raw patient records.

B

Differential Privacy (DP-SGD)

Per-sample gradient clipping and calibrated Gaussian noise prevent data reconstruction and inversion attacks.



Federated Averaging (FedAvg)

Central orchestrator aggregates weight updates across hospital nodes to build a high-accuracy global model.

C

Zero-Data Egress Compliance

Ensures verifiable compliance with HIPAA and DISHA by restricting patient data strictly to local hospital boundaries

D

FLOW OF SOLUTION



STEP 01

Local Data Training

Each hospital node (AIIMS, Tata, Apollo) trains a local diagnostic MLP on private clinical records inside its own isolated firewall.

STEP 02

Differential Privacy Injection

Client engines apply per-sample gradient clipping and calibrated Gaussian noise via DP-SGD to prevent reconstruction attacks.

STEP 03

Federated Aggregation

The central Flower orchestrator collects privacy-preserved model updates and executes FedAvg to compute global model weights.

STEP 04

Global Model Broadcast

The optimized global model is redistributed back to participating hospitals, iteratively improving cohort generalization each round.

STEP 05

Clinical Decision Support

Clinicians run high-confidence cancer inference locally while real-time dashboards track diagnostic accuracy and privacy budgets.

TECH STACK



AI & Diagnostics

Deep Learning Core

PyTorch & Scikit-Learn

3-layer diagnostic MLP trained on Wisconsin breast cancer features with local, zero-leakage scaling pipeline.



Federated Orchestration

Federated Network Flower (flwr) Framework

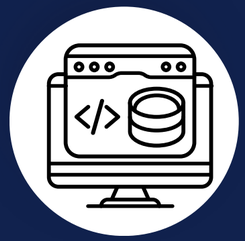
Decentralized FedAvg orchestration managing secure round updates across AIIMS, Tata, and Apollo hospital silos.



Privacy Engine

Differential Privacy Opacus (DP-SGD Engine)

Per-sample gradient clipping and calibrated Gaussian noise injection to mathematically guarantee privacy under DISHA/HIPAA.



Dashboard & Delivery

Clinical Dashboard Streamlit & Plotly

Real-time analytics portal tracking federated convergence, solo vs. federated accuracy gains, and live clinical inference.

UNIQUE SELLING PROPOSITION (USP)

01



Zero-Data-Egress Architecture

Trains diagnostic models locally inside hospital firewalls to eliminate patient data transfer risks.

02



Differential Privacy

Injects calibrated mathematical noise and gradient clipping to guarantee protection against model inversion attacks.

03



Cross-Institutional Generalization

Learns from multi-hospital cohorts (AIIMS, Tata, Apollo) without pooling or centralizing raw patient databases.

04



Solo vs. Federated Accuracy Gain

Consistently outperforms small, isolated hospital models by eliminating demographic bias and local overfitting.

05



Regulatory-First Compliance

Full alignment with HIPAA, GDPR, and DISHA healthcare data mandates via mathematical privacy guarantees.

06



Real-Time Clinical Decision Support

Delivers immediate malignancy predictions and live convergence tracking through an interactive clinical dashboard.

FEASIBILITY & COMPETITORS

FEASIBILITY

01 Decentralized Federated Learning

02 Differential Privacy (DP-SGD)

03 Multi-Hospital Generalization

04 Regulatory Compliance by Design

COMPETITORS

01 Centralized Medical Data Lakes

02 Isolated Hospital AI Models

03 Raw Gradient / Data Exposure

04 High Regulatory Risk